



COE-486 – Computer Vision Semester Project

The semester project is designed to assess your understanding of the course material and your ability to apply computer vision techniques to real-world problems. Working in **groups of 3** is expected. Select a project that is sufficiently scoped for completion within a 4-week timeframe. You may select one of the following options:

1. **Reproduce and Adapt:** Select a state-of-the-art research paper from a recent computer vision conference (see Appendix A). Publicly available project code for most of the papers can be accessed from [here](#). Apply the method to a new application domain (i.e. different dataset) and evaluate its performance.
2. **Extend and Innovate:** Choose a state-of-the-art research paper from a recent computer vision conference (see Appendix A). Utilize the publicly available project code and propose an extension to the method. Evaluate its performance against the original approach.
3. **Solve a Local Problem:** Identify a local computer vision problem relevant to AUS or UAE and develop a solution and evaluate its performance.
4. **Ensemble Learning:** Combine multiple computer vision methods to enhance the accuracy of a specific task. Implement the ensemble approach and evaluate its performance against individual models.

Your project must include all of the following steps:

- Data Indigestion: data cleaning, and preprocessing to make the data suitable for training.
- Data augmentation: artificially increasing the size of the training dataset by applying various transformations to the existing images.
- Deep learning model: deciding whether to leverage pre-existing deep learning models or create a new one.
- Performance metrics for evaluation: defining appropriate performance metrics for evaluating the performance.

Project Deliverables:

Final project report around 8 pages. Use latex template for the report <https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhncsfqn>

The project report must contain the following

- Abstract
- Introduction (Problem Identification)
- Literature Review/Background
- Contribution: whats the contribution, why should it take 1 month to work on it. Mention for each week what steps of the project were worked on. It should not include the literature review, only the things lrealted to the project, e.g methodology results error analusis.
- Methodology
- Results, training curves, accuracy curves etc

- Discussion

Submit code to GitHub. Make sure the **GitHub Readme** explains how to run the code step by step for both training and testing cases

Grading Rubric:

Criteria	Description	Weightage
Problem Identification and literature review	• problem complexity and relevance • clear problem statement and justification • Comprehensive literature review	15%
Methodology	• justification of Computer vision techniques applied • quality of implementation and code of GitHub • quality of the readme file • Innovative approach	30%
Result and analysis	• appropriate performance metrics • Comprehensive comparative evaluation with existing methods or between approaches	25%
Organization and structure of report	• Clear delivery of key findings, methodology, results and discussion	20%
Demo	• Quality of the demo	10%

Important Dates:

Submit Project Title, Abstract, Introduction, Literature Review): **5th April 2026**
 Submit Final Project Report, Demo, GitHub repo: **3rd May 2026**

Appendix A:

List of Computer Vision Conferences:

CVPR
 ICCV
 ECCV
 ACCV
 BMVC
 ICML
 NIPS
 AAAI
 MICCAI
 ICPR